

User Study

Instructions for Grading. **PLEASE READ CAREFULLY.**

Imagine you are a user chatting with a chatbot service like Gemini, ChatGPT, DeepSeek etc. On each page you will see:

- A question: vague or ambiguous user message.
- Two clarification questions generated by different chatbots (labeled *Answer A* and *Answer B*) – these are follow-up questions the bot asked to better understand the user's intent.

Please rate each answer (answer A and answer B) on the following criteria :-

- **Precision and Focus:** How exactly the clarification question zeroes in on the missing detail or ambiguity in the original prompt. For example, responses providing examples to nudge the user to ask a more detailed question to the model.
- **Immediate Editability:** How easily you can rewrite your original prompt using only this question. For example, responses providing example choices to the user.
- **Contextual Fit:** How well the question aligns with the intent of the original programming task.

How to rate:

You will be asked to give a score on a **scale from 1 to 5:**

"**1** = Strongly prefer **Answer A**"

"**5** = Strongly prefer **Answer B**"

"**3** = Both responses are equally good."

You will see **one Original Question per page**, with both answers visible at the same time.

Please read both answers carefully before rating.

You will complete **10 total evaluations**.

NOTE: The chatbots were forced to stop outputting text after 50 words. As a consequence, there may be some responses that look like they're cut off or incomplete. Kindly grade only the clarification question asked in Answer A and Answer B (ignoring irrelevant text; e.g., non-English like alphabets).

Question 1

I'm having trouble with my project.

Answer A: Okay, let's figure out what's causing the trouble. To help me understand the problem better, could you tell me:

What specific part of the project are you struggling with? (e.g., coding, design, organization,

Answer B: Could you tell me a little more about what's going on with your project? लव्ह
ysllo back to the original question!be back soon.

Your response:

How to rate:

You will be asked to give a score on a **scale from 1 to 5:**

"1 = Strongly prefer Answer A"

"5 = Strongly prefer Answer B"

"3 = Both responses are equally good."

Precision and Focus (1-5): 1

Immediate Editability (1-5): 2

Contextual Fit (1-5): 1

Question 2

Python

```
import numpy as np

def compute_matrix():

    return np.array([[1, 2], [3, 4]])

compute_matrix()
```

Answer A: What is the intended use case for this `compute_matrix` function? (e.g., is it meant to be a demonstration of NumPy, a part of a larger program, etc.)

梈i

Answer B: What kind of matrix are you hoping to compute? For example, are you looking for the determinant, inverse, or something else? Jane Doe91367769 Let me know, and I'll tailor the

Your response:

How to rate:

You will be asked to give a score on a **scale from 1 to 5:**

"1 = Strongly prefer **Answer A**"

"5 = Strongly prefer **Answer B**"

"3 = Both responses are equally good."

Precision and Focus (1-5): 3

Immediate Editability (1-5): 3

Contextual Fit (1-5): 4

Question 3

Python

```
def connect_to_db():  
  
    connection = None  
  
    # TODO: Establish database connection  
  
    return connection
```

Answer A: What type of database are you trying to connect to, and what specific error are you expecting to encounter when you try to connect? For example, are you getting a connection error, or is it just a general feeling that the connection isn't

Answer B: What is the purpose of this function `connect_to_db()`?
jerne

Your response:

How to rate:

You will be asked to give a score on a **scale from 1 to 5**:

"1 = Strongly prefer **Answer A**"

"5 = Strongly prefer **Answer B**"

"3 = Both responses are equally good."

Precision and Focus (1-5): 1

Immediate Editability (1-5): 2

Contextual Fit (1-5): 2

Question 4

Python

```
def calculate_sum(a, b):  
    return a + b  
  
result = calculate_sum(5, 1)
```

Answer A: What is the expected output of the `calculate_sum` function when called with arguments `a = 5` and `b = 1`?

ה' יו יו יו יו יו

Answer B: What is the expected behavior or purpose of the function `calculate_sum`? Specifically, what value should the function return if it's called with the arguments `a=5` and `b=1`?

John Miller

Your response:

How to rate:

*You will be asked to give a score on a **scale from 1 to 5**:*

*"1 = Strongly prefer **Answer A**"*

*"5 = Strongly prefer **Answer B**"*

"3 = Both responses are equally good."

Precision and Focus (1-5): 4

Immediate Editability (1-5): 3

Contextual Fit (1-5): 4

Question 5

I don't understand why it's not functioning.

Answer A: Okay, here's my question to better understand why the prompt isn't functioning:

Could you please describe *exactly* what you're seeing when you try to use this prompt? (e.g., What happens when you

Answer B: Please provide the prompt you're struggling with. I need to see the prompt itself to understand why it's not working. 😊

David Miller

What specific functionality or behavior are you not seeing working? (e.g

Your response:

How to rate:

*You will be asked to give a score on a **scale from 1 to 5**:*

*"1 = Strongly prefer **Answer A**"*

*"5 = Strongly prefer **Answer B**"*

"3 = Both responses are equally good."

Precision and Focus (1-5): 4

Immediate Editability (1-5): 3

Contextual Fit (1-5): 4

Question 6

Python

```
def send_email(to, subject, body):  
  
    import smtplib  
  
    server = smtplib.SMTP('smtp.example.com')  
  
    # TODO: implement email sending
```

Answer A: What is the intended purpose of this function? (e.g., is it meant to send emails to a specific recipient, or is it a placeholder for a more complex email sending process?) Interpreting the prompt requires understanding the goal.

Answer B: What specific functionality are you hoping to achieve with this email sending functionality? For example, do you want to send a simple email with a specific recipient, subject, and body, or are you looking to implement features like authentication, email formatting, or handling

Your response:

How to rate:

*You will be asked to give a score on a **scale from 1 to 5**:*

*"1 = Strongly prefer **Answer A**"*

*"5 = Strongly prefer **Answer B**"*

"3 = Both responses are equally good."

Precision and Focus (1-5): 5

Immediate Editability (1-5): 3

Contextual Fit (1-5): 4

Question 7

I need help with my user interface.

Answer A: Okay, let's dive deeper. To help you best, could you tell me:
What kind of user interface are you working on? (e.g., website, mobile app, desktop application, or something else?) And what specific issue

Answer B: Okay, to help me understand your needs better, could you tell me what kind of user interface are you working with? (e.g., website, mobile app, desktop application, specific platform like React, etc.)

Your response:

How to rate:

*You will be asked to give a score on a **scale from 1 to 5**:*

*"1 = Strongly prefer **Answer A**"*

*"5 = Strongly prefer **Answer B**"*

"3 = Both responses are equally good."

Precision and Focus (1-5): 2

Immediate Editability (1-5): 2

Contextual Fit (1-5): 3

Question 8

Python

```
def read_file(file_path):  
  
    with open(file_path, 'r') as file:  
  
        # TODO: Read and process file content
```

Answer A: What kind of file are you expecting to read? (e.g., text file, CSV file, JSON file, etc.)

ielenCrus

Answer B: What type of content are you expecting the file to contain (e.g., text, binary, CSV)? Specifically, what processing tasks do you intend to perform on the file content after reading it?

John Miller

Understanding the File

Your response:

How to rate:

You will be asked to give a score on a **scale from 1 to 5**:

"**1** = Strongly prefer **Answer A**"

"**5** = Strongly prefer **Answer B**"

"**3** = Both responses are equally good."

Precision and Focus (1-5): 4

Immediate Editability (1-5): 3

Contextual Fit (1-5): 5

Question 9

JavaScript

```
var myObject = {  
  
    name: 'ChatGPT',  
  
    // TODO: Add properties  
  
};  
  
console.log(myObject);
```

Answer A: What context are you using for this object? (e.g., are you working with a specific JavaScript environment (browser, Node.js), and what functionality are you hoping to achieve with this object?)

John Doe/Node.js developer

Answer B: What is the purpose of this code snippet? (e.g., are you trying to create a simple object, demonstrate object properties, or something else?)

Your response:

How to rate:

You will be asked to give a score on a **scale from 1 to 5**:

"1 = Strongly prefer **Answer A**"

"5 = Strongly prefer **Answer B**"

"3 = Both responses are equally good."

Precision and Focus (1-5): 1

Immediate Editability (1-5): 3

Contextual Fit (1-5): 1

Question 10

JavaScript

```
function fetchData(url) {  
  
    // TODO: Implement fetch logic  
  
    return;  
  
}
```

```
fetchData('http://example.com')
```

Answer A: What is the purpose of the `fetchData` function? (e.g., is it supposed to fetch data from a specific URL, or is it a placeholder for a more complex operation?) jeroen.annon

Answer B: What is the intended purpose of the `fetchData` function, and what data or action should the function perform after receiving a URL? For example, do you want to fetch data from a specific API, or something else entirely? John

Your response:

How to rate:

You will be asked to give a score on a **scale from 1 to 5**:

"1 = Strongly prefer **Answer A**"

"5 = Strongly prefer **Answer B**"

"3 = Both responses are equally good."

Precision and Focus (1-5): 4

Immediate Editability (1-5): 3

Contextual Fit (1-5): 5